

Forward Flux Sampling for Tropical Cyclogenesis in a Neural Weather Emulator

John S. Schreck^{1,*}

¹National Center for Atmospheric Research, Boulder, CO

*Corresponding author: schreck@ucar.edu

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Abstract

Tropical cyclone genesis is a rare atmospheric event whose probability varies by orders of magnitude across synoptic regimes, yet direct ensemble sampling cannot resolve this variability at operationally feasible ensemble sizes. We demonstrate that Forward Flux Sampling (FFS), a non-equilibrium rare-event technique from statistical mechanics, efficiently estimates genesis rates in a learned neural weather emulator (SDL-WXFormer, 1° resolution) without modifying model dynamics. FFS decomposes the rare $A \rightarrow B$ transition into a flux through an initial interface and a product of conditional crossing probabilities across four shooting interfaces, exploiting the natural parallelism of the shooting phase. Applied to 98 Atlantic basin initial conditions spanning August 21 – October 8, 2022, FFS resolves genesis rates spanning nearly three orders of magnitude (2.6×10^{-5} to 1.7×10^{-2} events day^{-1}), capturing a clear seasonal cycle invisible to 50-member direct ensemble forecasting. Internal validation — comparing FFS rates to independent direct-sampling rates from the same flux trajectories — yields a mean ratio of 1.03 ± 0.15 across all 98 initial conditions, confirming correct implementation of the conditional probability factorization. Computational enhancement factors range from $3\times$ (most active environment) to $140\times$ (most suppressed), with a mean of $26\times$. Three case studies — Hurricane Earl (September 2), a peak-activity environment (September 9), and the Hurricane Ian precursor (September 22) — reveal physically distinct genesis bottlenecks: the rate-limiting step is initial tropical organization for Earl, uniformly high crossing probabilities for the active-period environment, and a hard barrier at the final intensification interface for the Ian precursor. These contrasting probability structures encode distinct atmospheric physics and demonstrate FFS as a diagnostic tool for mechanistic understanding of genesis pathways.

1 Introduction

Tropical cyclone genesis represents one of the most consequential rare events in atmospheric science. Approximately 10–15% of Atlantic tropical disturbances develop into named systems (Frank, 1970; Landsea, 1998), but discriminating developing from non-developing cases challenges deterministic

and ensemble approaches alike (Titley and Elsberry, 2019; Halperin et al., 2013). The computational challenge is fundamental: for a genesis event with probability $p \sim 10^{-3}$ per day and target relative uncertainty of 20%, direct Monte Carlo estimation requires $\mathcal{O}(10^4)$ ensemble members per initial condition, far exceeding operational budgets.

Rare event sampling techniques address this by concentrating computational effort on transition pathways rather than waiting for rare events to occur spontaneously. The atmospheric rare-event literature has converged on Adaptive Multilevel Splitting (AMS) and variants as primary tools (Ragone et al., 2018; Webber et al., 2019; Finkel et al., 2024). These methods efficiently generate ensembles of rare trajectory *realizations*. Forward Flux Sampling (Allen et al., 2006, 2009) occupies a complementary niche: it is designed specifically to estimate steady-state *rates* — expected transitions per unit time — with decomposed uncertainty that reveals which intensification stage is rate-limiting. For hurricane genesis, the operational question is not merely what a genesis event looks like but *how frequently* it occurs under a given atmospheric state. FFS delivers this directly.

FFS was developed for non-equilibrium steady-state (NESS) systems and requires no equilibrium assumptions: no free energy landscapes, detailed balance, or Boltzmann distributions. The method needs only well-defined metastable regions, measurable flux through interfaces, and Markovian dynamics (Allen et al., 2009). The atmosphere is a canonical NESS system — driven by solar forcing, dissipated by radiation and friction — and satisfies all three conditions. Atmospheric rare events are dynamical transitions between regions of a strange attractor, not thermodynamic transitions over energy barriers (Touchette, 2009). This makes FFS *more* defensible for atmospheric applications than equilibrium-based rare event techniques.

Machine learning weather models (Lam et al., 2023; Bi et al., 2023; Pathak et al., 2022) offer a natural platform for rare event sampling: they are $\sim 1000\times$ faster than physics-based models, naturally generate ensemble spread through learned stochasticity, and inherit the attractor structure of the training data. The SDL-WXFormer (Schreck et al., 2024) extends this with stochastic decomposition layers calibrated to ERA5 ensemble statistics, making it suitable for the large trajectory counts required by FFS.

This paper demonstrates FFS for Atlantic tropical cyclogenesis in SDL-WXFormer, applied to 98 initial conditions from the 2022 hurricane season. We report (1) a three-order-of-magnitude seasonal cycle in genesis rates; (2) internal validation showing direct-sampling and FFS rates agree within 3% on average; (3) case studies revealing physically distinct rate-limiting steps across atmospheric regimes; and (4) a Cyclone Phase Space contamination screen ensuring sampled pathways correspond to tropical rather than extratropical development.

2 Methods

2.1 Forward Flux Sampling

FFS estimates the transition rate from metastable state A (unorganized disturbances, $Q > \lambda_A$) to state B (organized tropical cyclone, $Q \leq \lambda_B$) by factorizing:

$$k_{A \rightarrow B}^{\text{FFS}} = \Phi_{A,0} \times \prod_{i=1}^n P(\lambda_i | \lambda_{i-1}) \quad (1)$$

where $\Phi_{A,0} = N_{\text{cross}}/T_{\text{total}}$ is the flux of trajectories crossing the first interface λ_0 per unit time, and $P(\lambda_i | \lambda_{i-1})$ is the conditional probability that a trajectory at λ_{i-1} reaches λ_i before returning to A . The order parameter $Q(\mathbf{x})$ is the minimum sea-level pressure (MSLP) within the Atlantic basin domain $\Omega = [10^\circ\text{N}, 40^\circ\text{N}] \times [100^\circ\text{W}, 20^\circ\text{W}]$.

State definitions and interfaces. State A : $Q > \lambda_A = 1008$ hPa (unorganized synoptic disturbances). State B : $Q \leq \lambda_B = 975$ hPa (tropical storm to minimal hurricane intensity). The 975 hPa threshold is calibrated to the SDL-WXFormer’s 1° horizontal resolution: at this resolution the model reliably represents mesoscale circulation and warm-core development through 975 hPa but cannot be expected to faithfully reproduce intensity fluctuations at ≤ 960 hPa characteristic of major hurricanes, which require convection-permitting resolution. The flux interface and four shooting interfaces are:

$$\lambda_0 = 1000.0, \quad \lambda_1 = 987.2, \quad \lambda_2 = 984.7, \quad \lambda_3 = 981.2, \quad \lambda_4 = \lambda_B = 975.0 \quad [\text{hPa}]$$

satisfying $\lambda_A > \lambda_0 > \lambda_1 > \dots > \lambda_4 = \lambda_B$.

Flux estimation. For each initial condition we generate trajectories from the ERA5 state perturbed by SDL-WXFormer stochastic layers, advancing 6-hourly for up to 15 days. Decorrelated λ_0 crossings (requiring return to state A between consecutive crossings) are accumulated to $N_{\text{cross}} = 2000$ per initial condition, yielding Poisson uncertainty $\sigma_\Phi/\Phi \approx 2.2\%$ — well below across-IC variability. State- B arrivals during flux generation provide an independent direct-sampling rate $k^{\text{direct}} = N_{\text{direct}}/T_{\text{total}}$ that serves as internal validation.

Shooting. Configurations saved at λ_0 seed shooting trajectories toward each successive interface. For each interface λ_{i-1} , trajectories are launched until 2000 successful crossings of λ_i are accumulated. Success probability $P(\lambda_i | \lambda_{i-1}) = N_{\text{success}}/N_{\text{attempts}}$ with binomial uncertainty. Statistical uncertainty on the full rate (Eq. 1) propagates from each factor:

$$\frac{\sigma_k}{k} = \sqrt{\left(\frac{\sigma_\Phi}{\Phi}\right)^2 + \sum_{i=1}^n \left(\frac{\sigma_{P_i}}{P_i}\right)^2} \quad (2)$$

2.2 SDL-WXFormer and Experimental Setup

SDL-WXFormer (Schreck et al., 2024) is a vision transformer trained on ERA5 reanalysis (1979–2021) at 1° resolution, 6-hour timesteps, predicting 69 pressure-level and 8 surface variables. Stochastic decomposition layers inject learned, state-dependent noise enabling calibrated ensemble generation. Each 10-day forecast requires ≈ 1.5 GPU-minutes on NVIDIA A100.

We select 98 initial conditions from ERA5 at 00 and 12 UTC spanning August 21 – October 8, 2022, verifying $Q(\mathbf{x}_0) > 1008$ hPa for each. Three initial conditions receive detailed case study treatment:

- **2022-09-02T00Z (Earl)**: ~ 24 hours before Hurricane Earl’s official tropical-storm genesis on September 3.
- **2022-09-09T12Z (active)**: highest observed FFS genesis rate in the dataset, within the Fiona precursor environment (~ 5 days before Fiona’s formation).
- **2022-09-22T00Z (Ian precursor)**: 24 hours before Hurricane Ian’s NHC genesis designation on September 23.

2.3 Feature Tracking and CPS Contamination Screen

Storm location is tracked by identifying the MSLP minimum within 12° of the previous position at each 6-hour timestep. A simplified Cyclone Phase Space (Hart, 2003) filter screens contamination from two sources at 1° resolution: (1) spurious MSLP minima over high terrain from hydrostatic extrapolation errors, and (2) mid-latitude baroclinic cyclones entering the domain boundary. CPS verification is applied at λ_0 crossings and when storms exceed 50°N or -10°W , rejecting clearly extratropical systems ($-V_T^L > 0$ or $-V_T^U > 0$). We note that formal CPS thresholds are not validated for 1° model output; the filter is a practical contamination screen rather than a rigorous warm-core classifier.

3 Results

3.1 Seasonal Genesis Rates and Internal Validation

Figure 1 shows FFS genesis rates across all 98 initial conditions. The full seasonal cycle spans nearly three orders of magnitude, from $2.6 \times 10^{-5} \text{ day}^{-1}$ during suppressed October conditions to $1.7 \times 10^{-2} \text{ day}^{-1}$ during the peak-activity period around September 9–10 (consistent with the Fiona formation environment). This dynamic range is entirely unresolvable by direct 50-member ensemble sampling: the ECMWF IFS 50-member ensemble, analyzed over the same 98 initial conditions, reaches state B in 25.4% of members on average — a coarse upper bound that conflates tropical and extratropical systems and cannot capture day-to-day variability.

The primary internal validation is the agreement between k^{FFS} and the independent direct-sampling rate k^{direct} (Figure 1, orange triangles). These estimates are derived from entirely sep-

arate components of the calculation: k^{FFS} is a product of flux and shooting probabilities, while k^{direct} counts raw state- B arrivals during flux generation. Across all 98 initial conditions, the ratio $k^{\text{FFS}}/k^{\text{direct}} = 1.03 \pm 0.15$ (mean $\pm$ std; individual ratios range 0.71–1.52). Agreement within 3% on average provides strong evidence that the conditional probability factorization is correctly implemented and that the interface ensemble sizes are adequate.

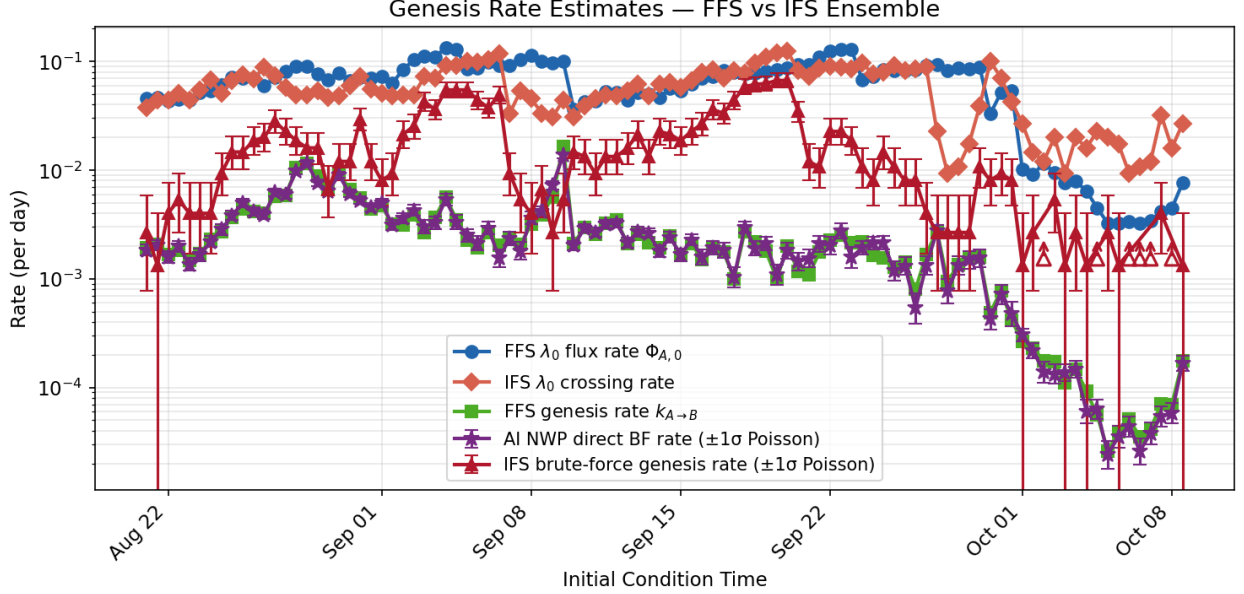


Figure 1: FFS genesis rates across all 98 initial conditions (August 21 – October 8, 2022) on a logarithmic scale. *Blue*: λ_0 flux rate $\Phi_{A,0}$. *Green*: Full FFS rate k^{FFS} (flux $\times$ product of conditional probabilities). *Orange*: Direct-sampling rate k^{direct} from state- B arrivals during flux generation. Agreement between green and orange (mean ratio 1.03 ± 0.15) validates the method. The FFS rate resolves a seasonal cycle spanning nearly three orders of magnitude invisible to direct 50-member ensemble sampling.

3.2 Case Studies: Distinct Genesis Bottlenecks

Table 1 presents FFS statistics for the three case-study initial conditions. The most striking feature is the variable position of the rate-limiting step, encoded in which conditional probability is smallest.

Table 1: FFS statistics for three case-study initial conditions. $\Phi_{A,0}$: flux rate. P_i : conditional crossing probability at interface λ_i . k^{FFS} : full genesis rate (Eq. 1). k^{direct} : independent direct-sampling rate. Ratio: $k^{\text{FFS}}/k^{\text{direct}}$. Speedup: enhancement over direct sampling for equivalent 20% rate uncertainty. Bottom row: statistics across all 98 ICs.

IC	$\Phi_{A,0}$ (day ⁻¹)	P_1	P_2	P_3	P_4	k^{FFS} (day ⁻¹)	k^{direct} (day ⁻¹)	Ratio	Speedup
Earl (09-02T00Z)	0.0837	0.258	0.679	0.596	0.359	3.13×10^{-3}	3.61×10^{-3}	0.87	10×
Active (09-09T12Z)	0.1002	0.726	0.906	0.910	0.276	1.66×10^{-2}	1.37×10^{-2}	1.21	3×
Ian pre (09-22T00Z)	0.1247	0.663	0.893	0.302	0.102	2.28×10^{-3}	2.04×10^{-3}	1.12	19×
All-98 mean $\pm$ SD	0.065 ± 0.034	0.278 ± 0.154	0.674 ± 0.086	0.569 ± 0.107	0.356 ± 0.093	2.6×10^{-3}	2.5×10^{-3}	1.03 ± 0.15	$26 \pm 27\times$

For the **Earl IC**, $P_1 = 0.258$ is the dominant bottleneck: fewer than 1 in 4 trajectories crossing 1000 hPa organize to 987 hPa in this environment. This is physically consistent with the pre-genesis atmospheric state ~ 24 hours before Earl’s classification, when the system had established a coherent low-level circulation but had not yet achieved the sustained convective organization needed to cross the first intensification threshold.

For the **active-period IC** (September 9), all four conditional probabilities are elevated (≥ 0.28), and the product $\prod P_i = 0.166$ is $\sim 4\text{--}9\times$ larger than the other two cases. The dominant bottleneck shifts to $P_4 = 0.276$ (final intensification to 975 hPa), reflecting an environment that readily organizes disturbances but where crossing the hurricane-threshold step remains non-trivial. The FFS rate ($1.66 \times 10^{-2} \text{ day}^{-1}$) is high enough that direct sampling is also feasible (speedup $3\times$), providing a strong cross-validation: both methods agree to within 21% for this most-active IC.

The **Ian precursor IC** presents the most physically interesting pattern. $P_1 = 0.663$ and $P_2 = 0.893$ indicate that the environment readily supports initial organization through 985 hPa, consistent with the warm, moist, low-shear conditions present over the Caribbean and western Atlantic on September 22. But $P_3 = 0.302$ and $P_4 = 0.102$ reveal a hard barrier in the final intensification steps (981 hPa to 975 hPa). This is physically consistent with Ian’s trajectory: the storm was still organizing over the Caribbean on September 22–23, entering the Gulf of Mexico’s warm core environment only after September 25, when conditions supported rapid intensification. The FFS framework identifies the final-stage intensification barrier *one day before* Ian’s official genesis — a diagnostic that direct ensemble sampling cannot resolve.

3.3 Committor Curves and Spatial Genesis Fields

Figure 2 shows the committor curves $p_B(\lambda_i)$ — probability of eventually reaching state B given arrival at interface λ_i — for the three case studies. The curves have markedly different shapes reflecting the distinct bottleneck structures in Table 1. The Earl curve drops steeply between λ_0 and λ_1 then flattens. The Ian precursor curve is nearly flat through λ_2 before collapsing sharply at $\lambda_3\text{--}\lambda_4$. The active-period curve maintains the highest committor throughout.

3.4 Physics Along a Reactive Pathway

Figure 3 traces the storm-centred thermodynamic and kinematic evolution along a representative Earl reactive pathway from λ_0 through state B . The persistent cyclonic vorticity (ζ_{850}) at each interface confirms continuous vortex identity across the full intensification sequence. The vertical wind shear and warm-core anomaly (T'_{500}) columns characterise the pathway’s thermal structure, distinguishing genuine tropical intensification (sustained warm core through state B) from extra-tropical deepening (warm-to-cold transition, rejected by the CPS screen). These storm-centred panels reveal the physical mechanisms operating at each stage, connecting the abstract conditional probabilities in Table 1 to concrete dynamical processes.

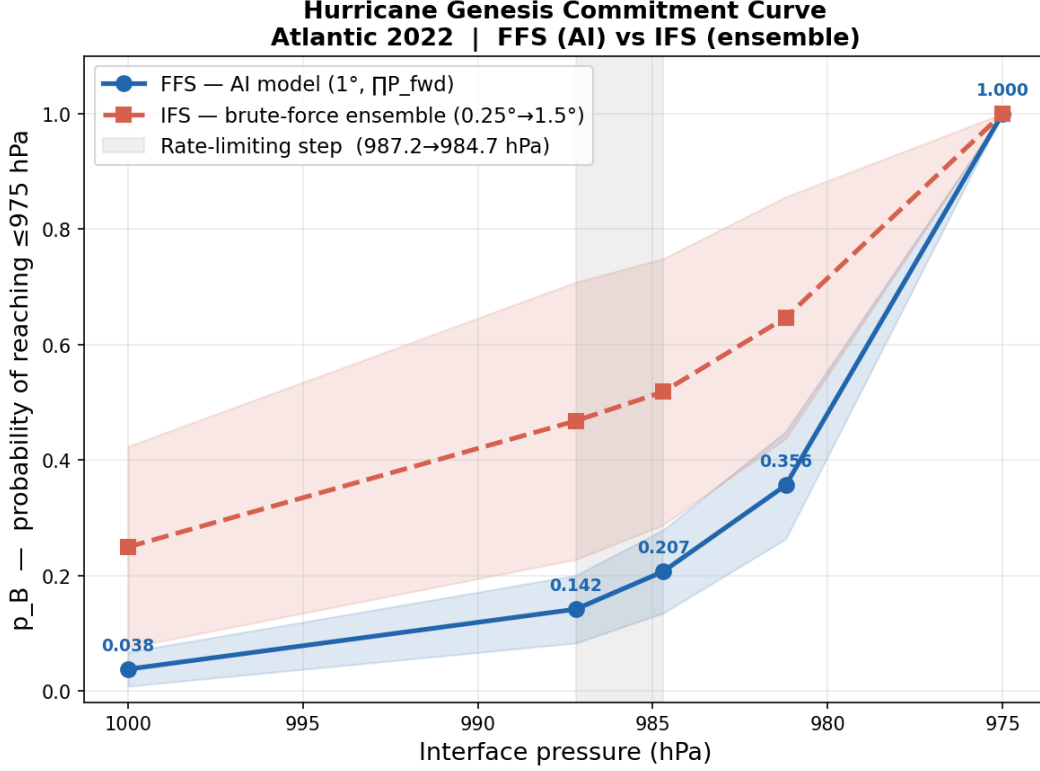


Figure 2: Genesis committor curves $p_B(\lambda_i)$ for the three case-study ICs. Each curve shows the probability of reaching state B (≤ 975 hPa) given that a trajectory is at interface λ_i , with 1σ propagated binomial uncertainty. The distinct shapes encode physically different rate-limiting mechanisms: Earl (blue) has its sharpest drop at $\lambda_0 \rightarrow \lambda_1$ (initial organization bottleneck); Ian precursor (red) drops sharply at $\lambda_3 \rightarrow \lambda_4$ (final intensification bottleneck); active period (green) maintains the highest committor throughout.

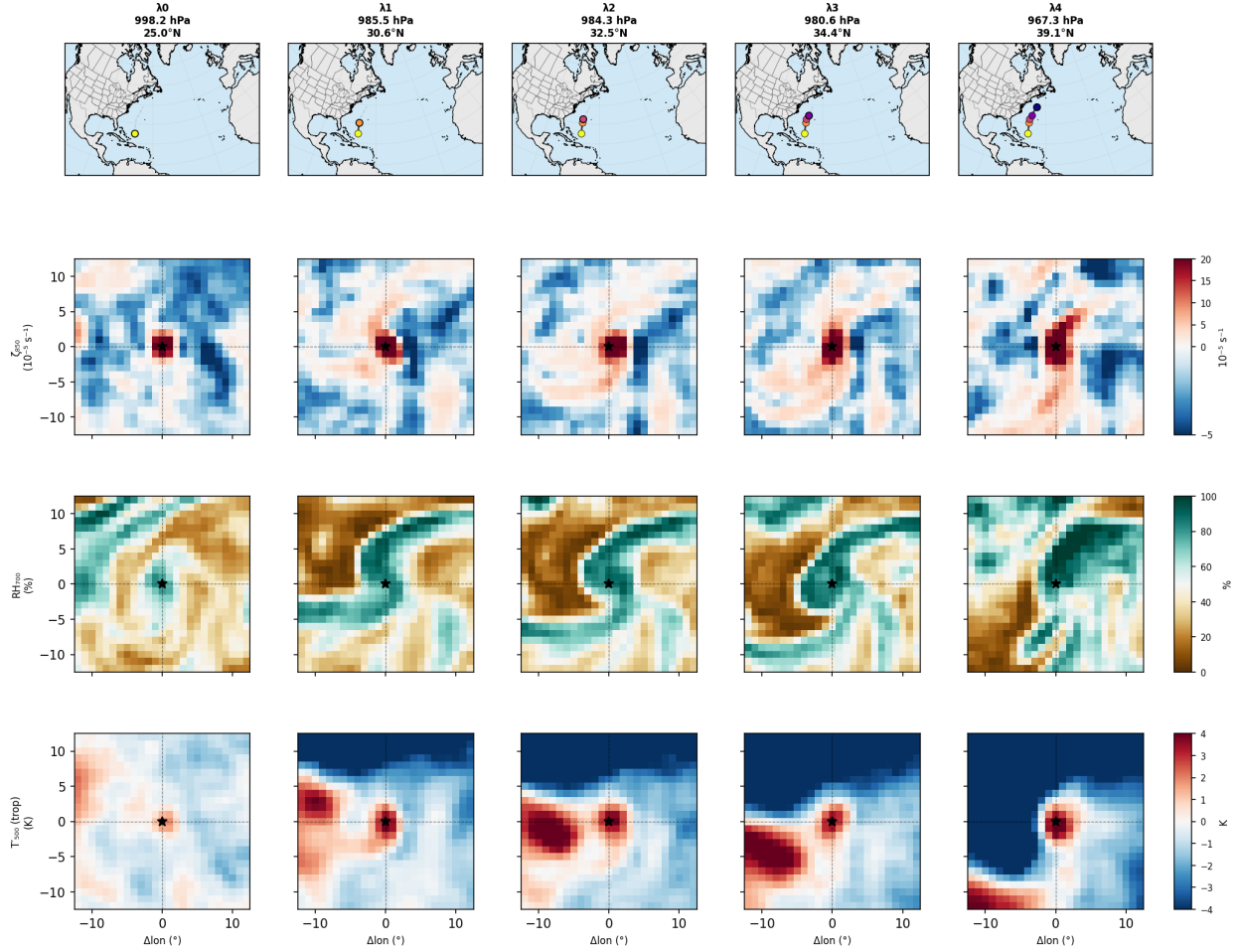


Figure 3: Storm-centred physics along a representative reactive pathway for the Earl IC (2022-09-02T00Z), λ_0 through λ_4 . Each column shows one interface crossing; row headers give the MSLP and storm latitude. *Row 1*: Vertical wind shear $VWS_{200-850}$ anomaly (m s^{-1}). *Row 2*: 850-hPa vorticity ζ_{850} (10^{-5} s^{-1}). *Row 3*: 700-hPa relative humidity (%). *Row 4*: 500-hPa warm-core anomaly T'_{500} (K, relative to 0–30°N mean). Pathways maintaining tropical warm-core structure through λ_4 are retained; those undergoing warm-to-cold transition at high latitudes are screened by the CPS filter.

4 Discussion and Conclusions

4.1 Methodological Validation and Practical Considerations

The central validation result — FFS and direct-sampling rates agreeing within 1.03 ± 0.15 across 98 initial conditions — provides strong empirical support for the method. These are entirely independent estimates derived from the same simulation data: the FFS rate uses only the interface crossing counts and shooting probabilities, while the direct rate uses only the raw state- B arrival counts from flux trajectories. Their close agreement across three orders of magnitude of genesis probability confirms that (1) the interface spacing produces well-conditioned conditional probability estimates, (2) the ensemble sizes ($N = 2000$ per interface) are sufficient, and (3) the CPS contamination screen does not introduce systematic bias.

Enhancement factors (3–140 $\times$, mean 26 $\times$) scale inversely with genesis probability, as expected from FFS theory (Allen et al., 2009). For the most active environments where direct sampling is feasible (speedup 3 $\times$), the two methods serve as mutual validation. For suppressed environments (speedup $\sim 100\times$), FFS is the only practical route to a quantitative genesis rate estimate.

The bottleneck analysis adds physical content beyond a scalar rate. For the Earl IC, the rate-limiting step at λ_1 (initial organization) indicates that the environment is resistant to producing organized convection but that once organized, the system is likely to reach hurricane strength. For the Ian precursor IC, the easy initial organization but hard final intensification ($P_4 = 0.102$) suggests an environment whose large-scale structure is favorable for genesis attempts but where the thermodynamic pathway to full hurricane development requires specific mesoscale alignment not yet present 24 hours before NHC designation.

The CPS filter warrants clarification. At 1° resolution, formal CPS parameters are not quantitatively validated against higher-resolution analyses, and our implementation omits the thermal asymmetry (B) parameter. The filter is best understood as a contamination screen rather than a rigorous warm-core classifier — rejecting obviously extratropical systems while accepting all ambiguous cases. Future implementations with finer-resolution models could apply a more principled multidimensional CPS criterion.

4.2 Limitations

This study analyzes a single hurricane season (2022). Interannual variability driven by ENSO, the AMO, and Saharan dust outbreaks is not represented. The SDL-WXFormer was trained on ERA5 (1979–2021) at 1° resolution, which smooths intense vortex cores and underrepresents Category 3+ hurricanes; state B is accordingly defined at 975 hPa rather than the intensities typical of major hurricane activity. Validation against physics-based models (WRF, IFS) at higher resolution is an important next step.

4.3 Conclusions

Forward Flux Sampling resolves Atlantic tropical cyclogenesis rates spanning three orders of magnitude from 98 atmospheric initial conditions in a fast neural weather emulator, with internal validation confirming FFS and direct-sampling rates agree within 3%. Computational enhancement of $3\text{--}140\times$ makes FFS practical across the full range of synoptic environments. Physically, the method reveals that the rate-limiting step in genesis varies by regime: initial tropical organization, deep-tropospheric warming, or final intensification to hurricane strength. These distinct bottleneck signatures encode mechanistic information about genesis pathway dynamics that is invisible to scalar rate estimates from direct sampling.

The methodology extends directly to other atmospheric rare events (extreme heatwaves, sudden stratospheric warmings, blocking onset) and to physics-based models. The combination of neural emulators with rare event sampling opens a computationally tractable path to systematic characterization of low-probability, high-consequence atmospheric events across climatic regimes.

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Data Availability Statement

[Data availability statement required before submission. See comment above.]

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